CS & IT ENGINEERING



Algorithms

Greedy Method

Lecture No.- 06



By- Aditya sir

Recap of Previous Lecture



- Topic
- Topic

Huffman Encoding

MCST

Topics to be Covered



- Topic
- Topic
- Topic

MCST



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.

Telegram



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

Applications of MCST ? → Optimization

1) Computer Networks:
   - Broadcasting
   - Multicasting

2) Digital electronics → Hardware designing.

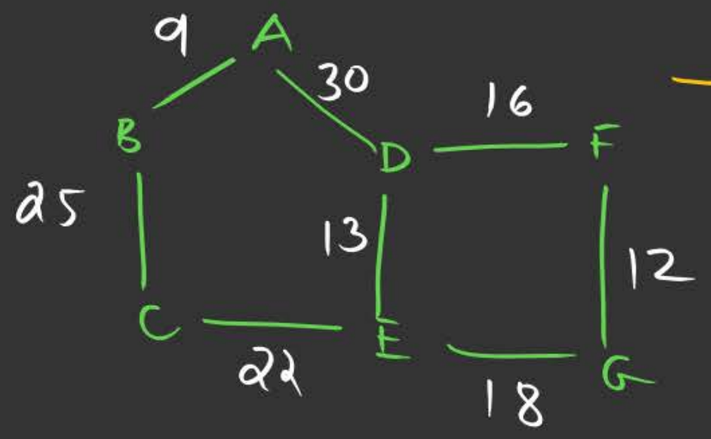
Cost of MCST
   - (unique)

Algorithms for MCST Construction :-

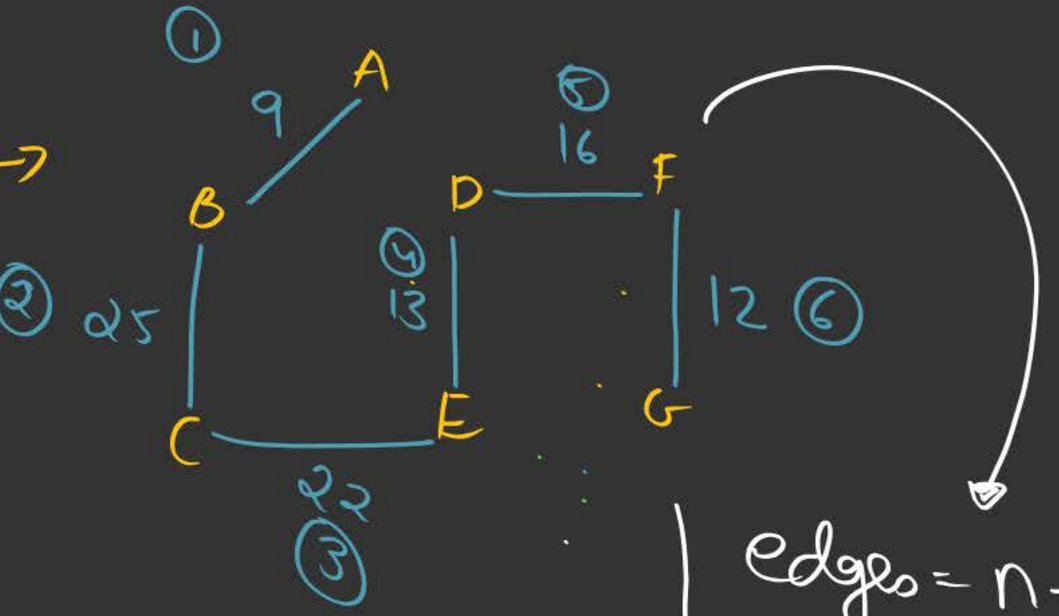
↳ 1) Prims Algo

2) Kruskal Algo

1) Prims Algo.



MCST



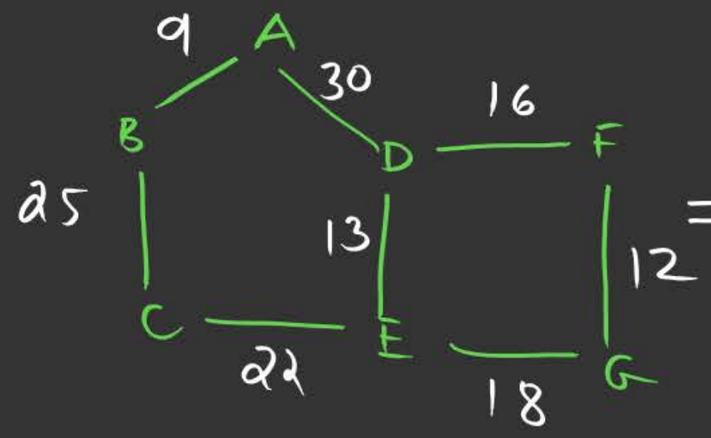
n=7

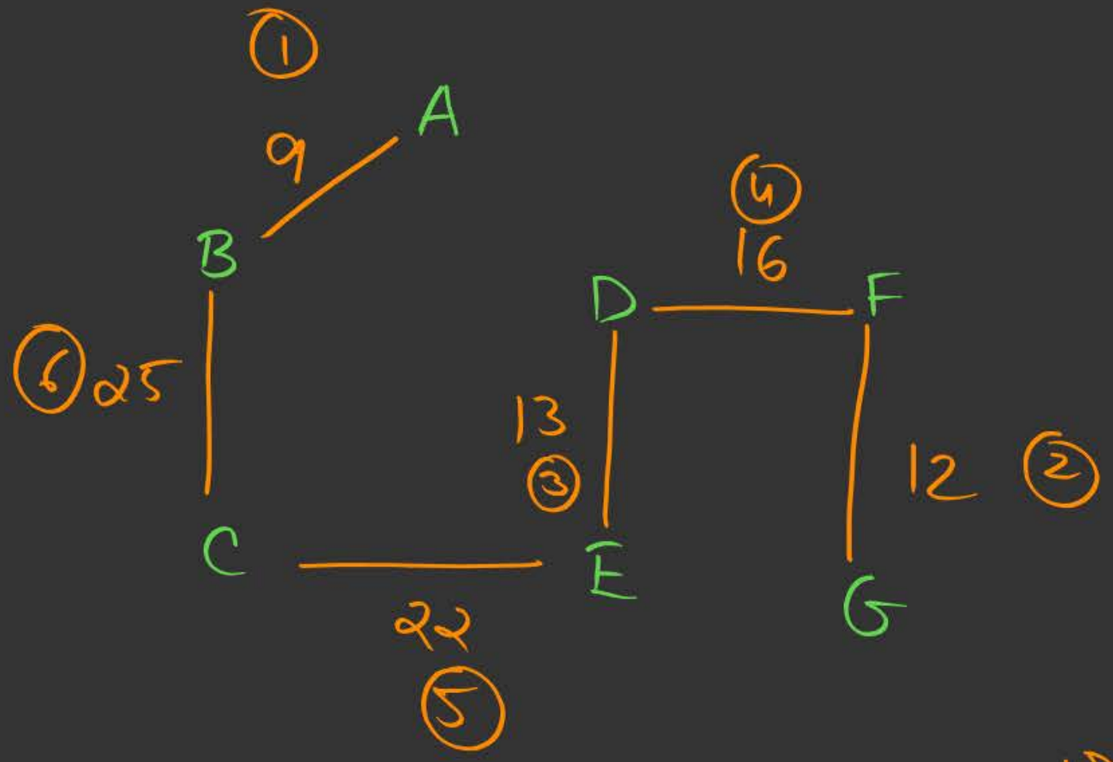
edges = n-1
= 7-1 = 6

Cost of MCST

= 9+25+22+13+16+12 = 97

1) Kruskal Algo :





{ Min Heap + Set }

Cost MCST
= 9+25+22+13+16+12
= 97

18 → $8R^3p$



Topic : Greedy Algorithm

For a given Graph : G(V, E), |V| = n, |E| = e

1) Prims Algo: TC = $O(n^2)$

2) Kruskal Algo: TC = O(e log e)

3) Prims Algo always maintains Tree Structure property at each step, whereas Kruskal may or may not.

4) Cost of Spanning Tree (MCST) by both the approaches will always be same.

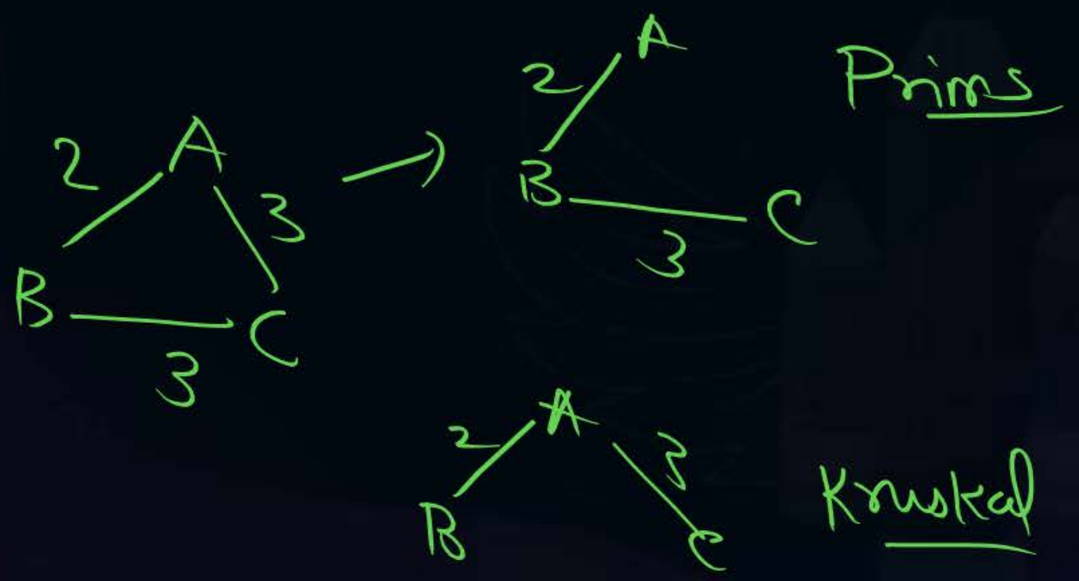
5) The tree structure of the MCST obtained by both may or may not be same.



Topic : Greedy Algorithm

a) If all the edges have unique/distinct cost, then the structure of MCST by both algo is also same.

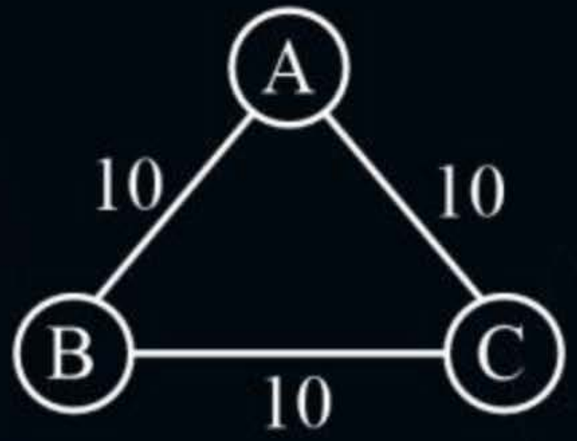
b) If there are duplicate weighted edges, then the structure of MCST may or may not be the same.





Topic : Greedy Algorithm

Example: n = 3, e = 3

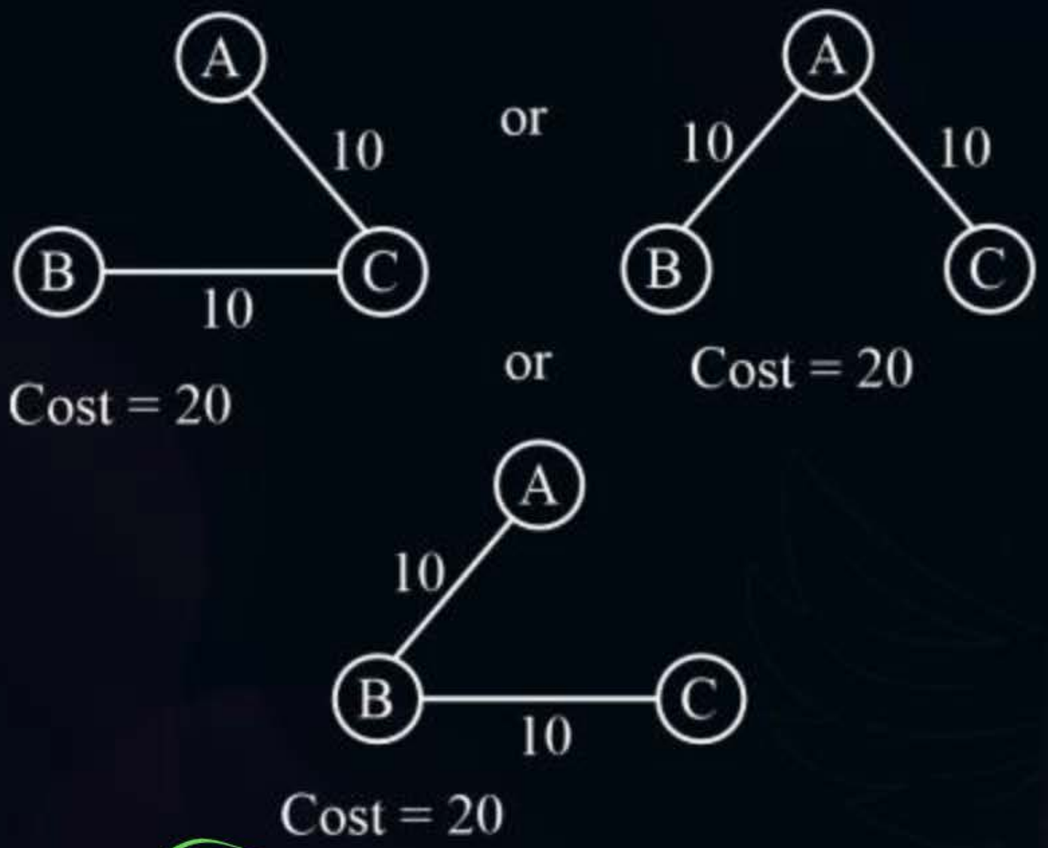


Cost (MCST) = 20



Topic : Greedy Algorithm

MCST:



- Hence, cost of MCST is always unique. But, there can be multiple MCSTs (as there are duplicate weighted edges)



Topic : Greedy Algorithm

Important:

#Q. If all the edges in the G(V, E) are equal (equi-weighted graph), then the cost of MCST = ?

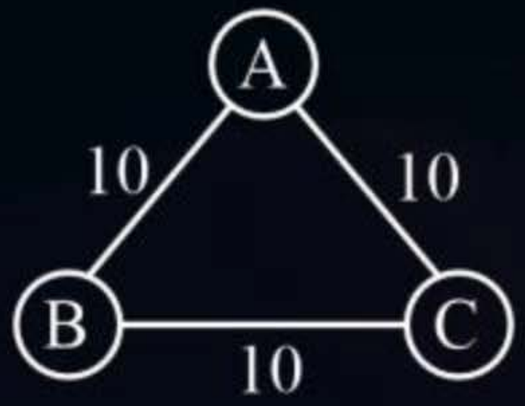
All C

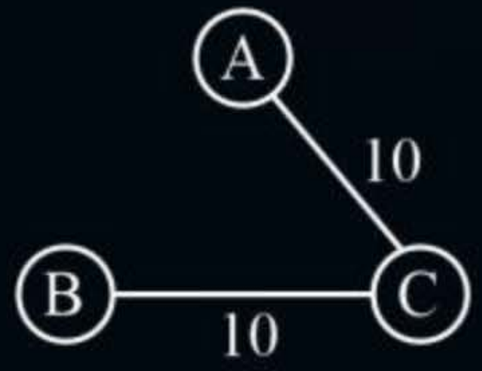
C*(n-1)



Topic : Greedy Algorithm

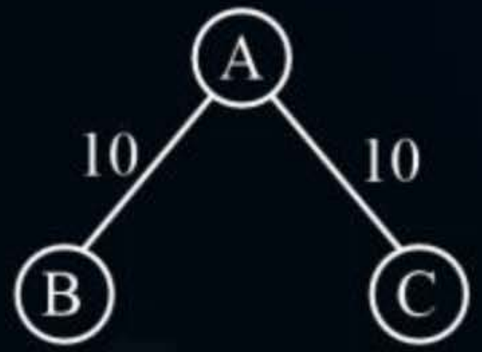
10, 10, 10





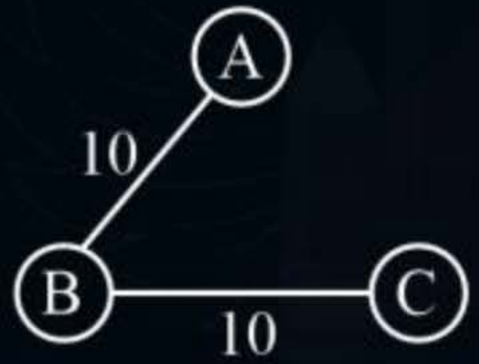
Cost = 20

or



Cost = 20

or



Cost = 20

C = 10
n = 3 ⇒

(3-1)×10
= 20



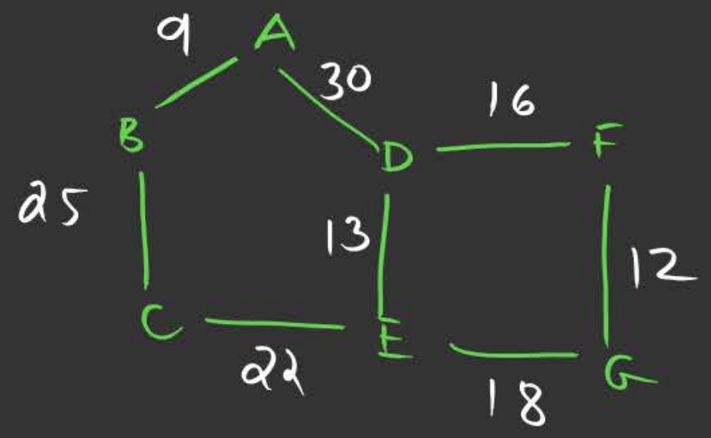
Topic : Greedy Algorithm

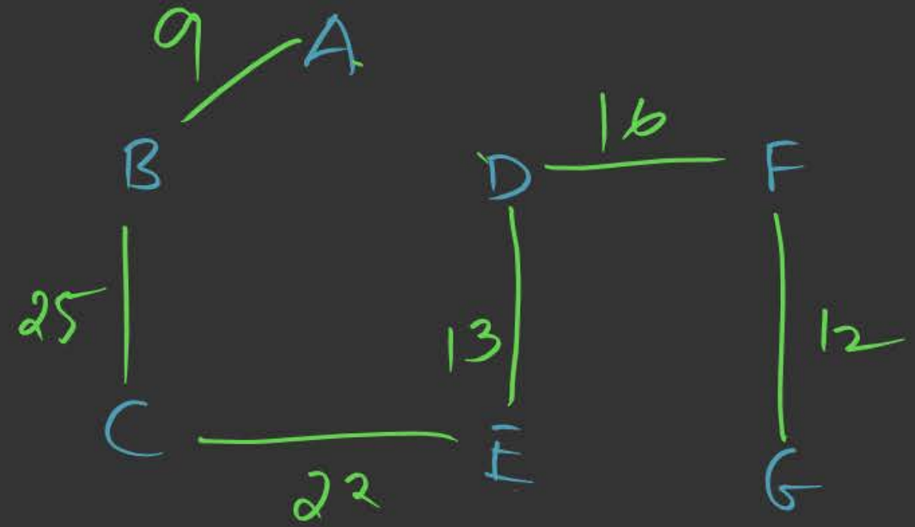
Dijkstra's MCST Construction Algorithm: (exam shortcut)

Steps:

1) Randomly Pick edges (consider all of them)
2) Only when cycle gets formed, remove the max cost edge from that cycle and continue until all edges are considered.

③ Dj bro





Cost of MCST =

$= 9 + 25 + 22 + 13 + 16 + 12 = 97$

[NAT]

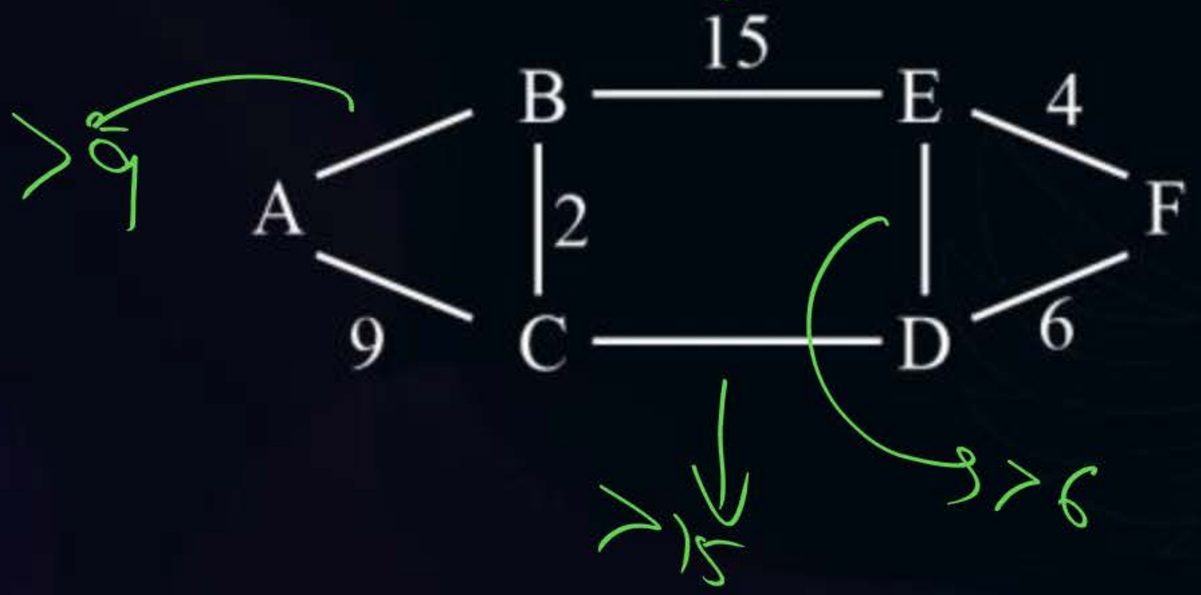
PYQ

#Q. Consider the following graph whose minimum cost spanning tree marked with edges values have a weight of 36.

Minimum possible cost of all the edges of the graph G is _____?

Assume that all the edges have distinct cost.

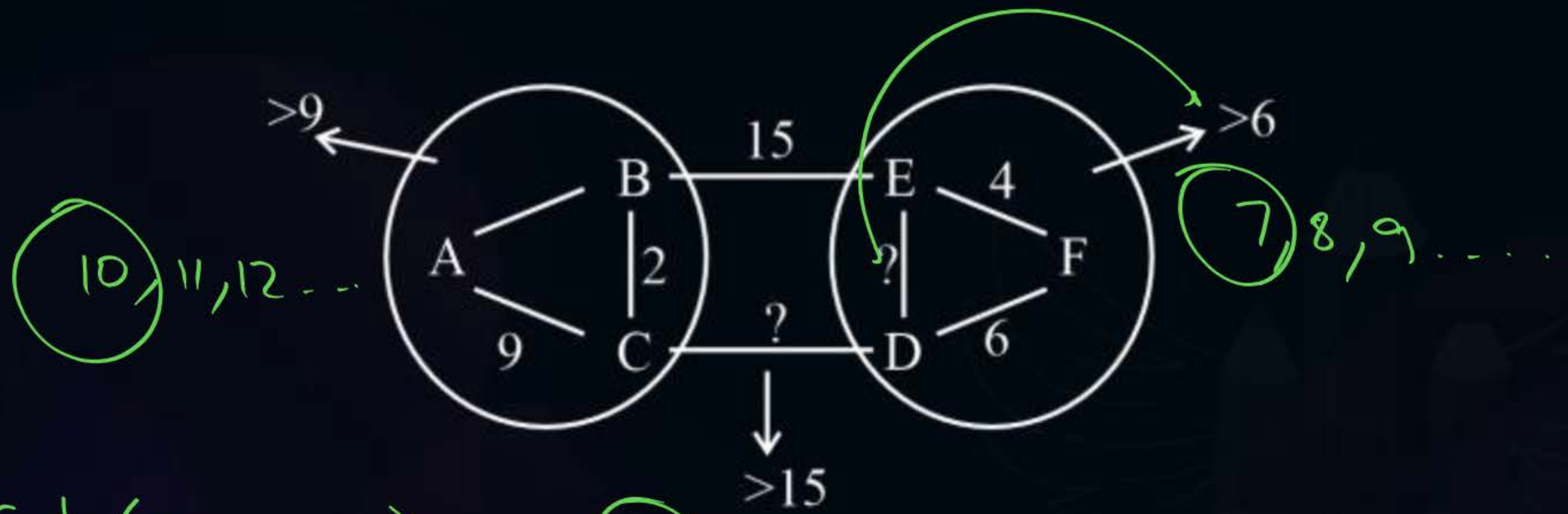
30.6%





Topic : Greedy Algorithm

As per Dijkstra MCST Approach:



Ans: $36 + (10 + 16 + 7)$

$= 36 + 33 = 69$ ✓

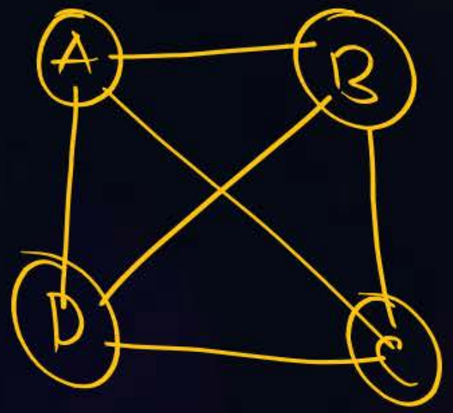
[NAT] PYQ

#Q. Let G be a complete undirected graph with 4 vertices and edge weights are {1, 2, 3, 4, 5, 6} The maximum possible weight that a minimum weight spanning tree can have is _____.

10-7%

6

$n = 4$



max

$MCST_1 \rightarrow C_1$

$MCST_2 \rightarrow C_2$

$MCST_3 \rightarrow C_3$

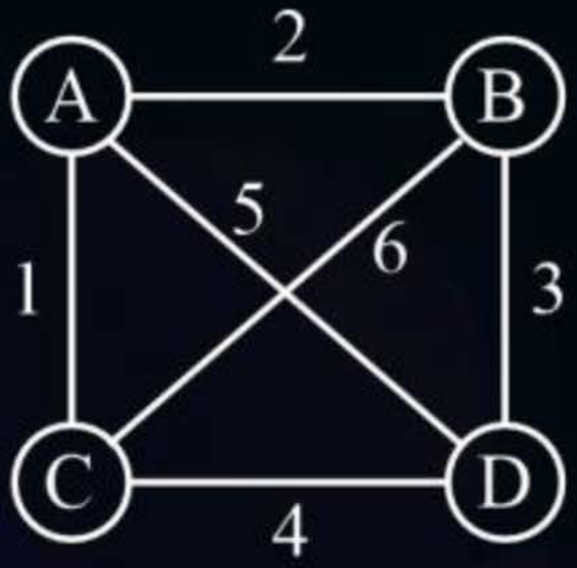
$MCST_K \rightarrow C_K$



Topic : Greedy Algorithm

Solution:

G1:



MCST

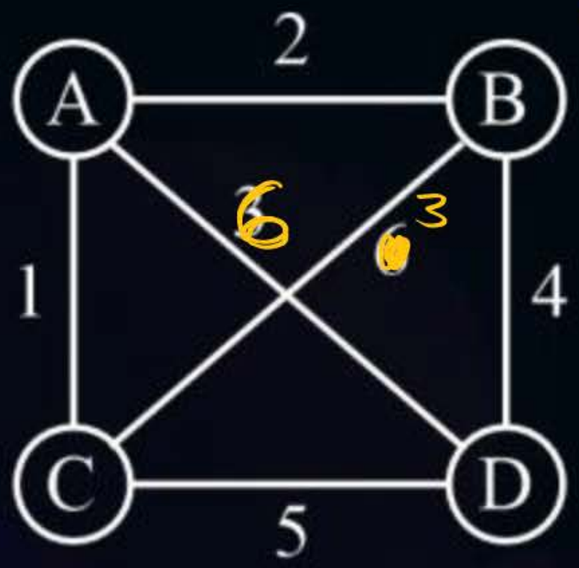


Cost = 1 + 2 + 3 = 6

Topic : Greedy Algorithm

valid MCST

G2:



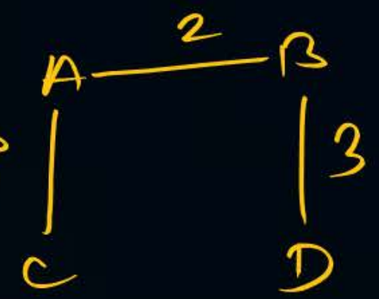
MCST →



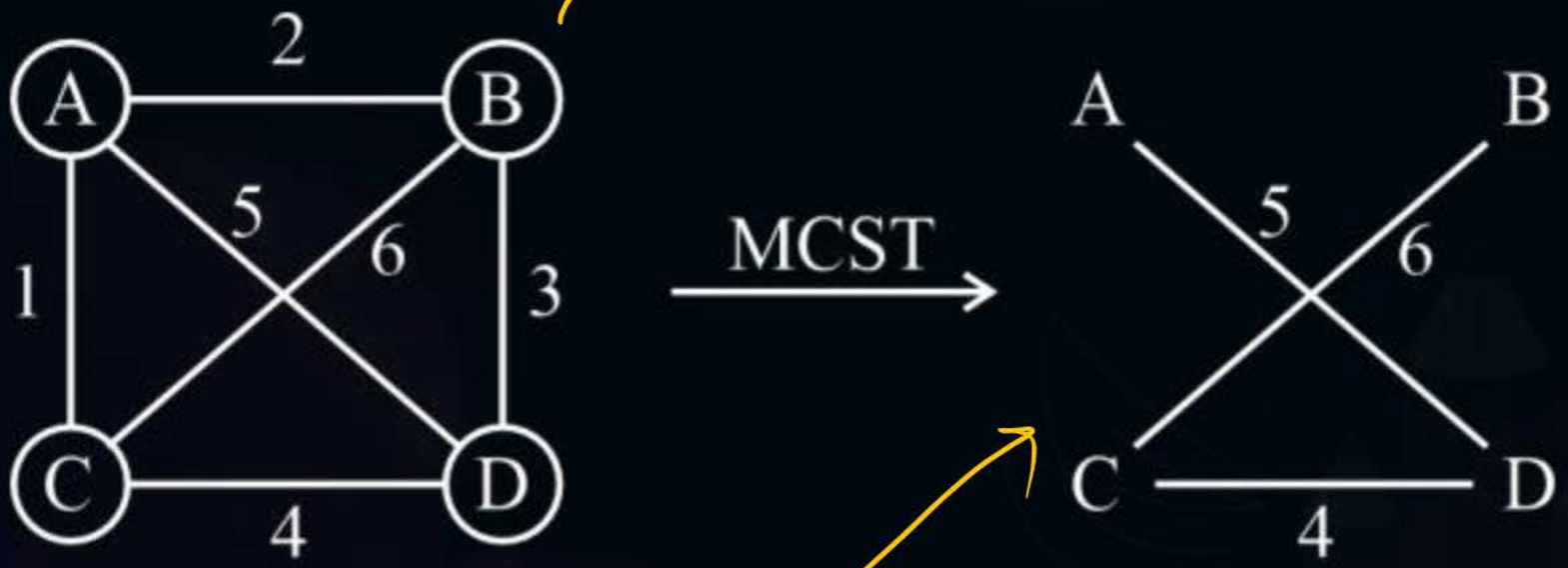
Cost = 1 + 2 + 4 = 7



Topic : Greedy Algorithm

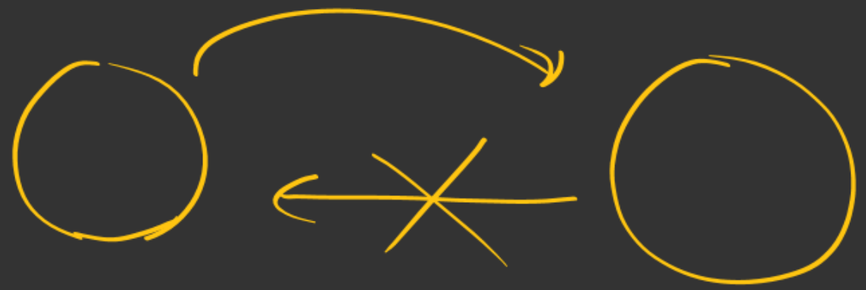


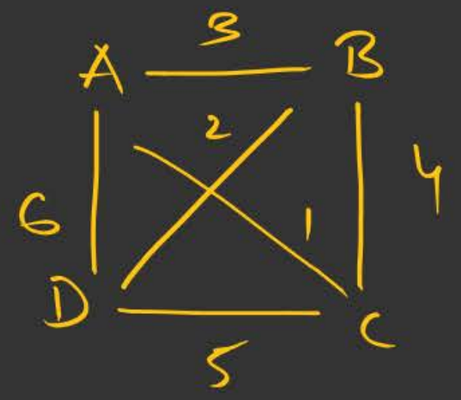
G4:

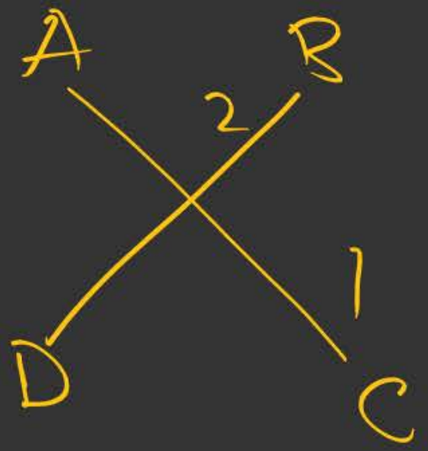


Cost = 6 + 5 + 4 = 15

- Not a valid MCST




A
3
B
2
6
4
1
D
C
5


A
B
2
1
D
C

[MCQ]

#Q. Consider a complete weighted graph with n vertices numbered V_1 to V_n. Two vertices V_i and V_j having edge between them has a cost value of $2\,|i - j|$. The weight of minimum cost spanning tree of such a graph is _____.

77%

n = 3

A) 2n → 2×3 = 6 ✗

B) 3n − 2 → 3×3 − 2 = 9 − 2 = 7 ✗

C) 2n − 2 → 2×3 − 2 = 6 − 2 = 4 ✓

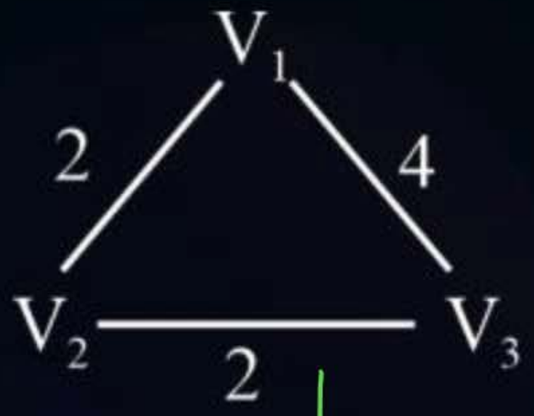
D) 2n − 1 → 2×3 − 1 = 6 − 1 = 5 ✗



Topic : Greedy Algorithm

Solution:

1)

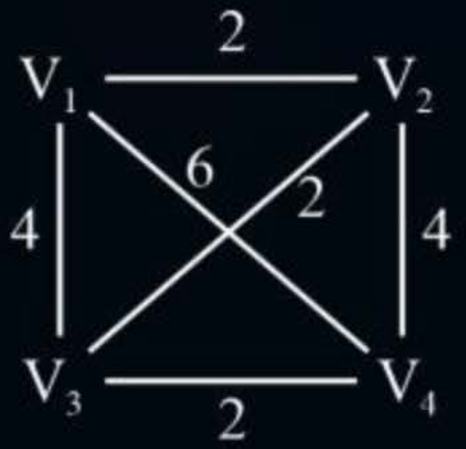


MCST



cost = 4

2)





Obs :-



$2*(i+1-i) = 2 \times 1 = \underline{2}$



Chain of Adjacent vertices

mcst → edges : $\underline{(n-1)}$

every edge will be of cost 2 $\Rightarrow$ Total cost of MST $= 2*(n-1) = \underline{2n-2}$

[NAT]

#Q. Let G a connected undirected graph of 100 vertices and 300 edges. The weight of a minimum spanning tree of G is 500. When the weight of each edge of G is increased by five, the weight of a minimum spanning tree becomes _____.

H.W



2 mins Summary

- Topic
- Topic
- Topic
- Topic
- Topic

MCST
- Prims
- Kruskal
- Dijkstra

} PYQs



THANK - YOU